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# German Celebrities and Political Communication on Social Media – Opinion Leadership and Endorsements

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**Abstract**

Celebrities play an increasingly important role in political communication in the United States, primarily operating through social media. This study examines whether comparable patterns can be observed in Germany and how celebrity involvement affects the performance of political content on Instagram. The analysis is based on two large-scale datasets comprising 330,562 posts by German and U.S. celebrities and 149,460 posts by German politicians. Using machine learning–based classification and inferential statistical models, four hypotheses are tested: (1) political communication by celebrities on social media differs between the United States and Germany; (2) the type of political representation used by German celebrities—issue-based, party-political, or personalized—has differential effects on the performance of political posts; (3) political and non-political celebrity posts generate different levels of engagement; and (4) posts by politicians benefit from visible support by celebrities.

The results show that political content accounts for only 1.7% of German celebrity posts, compared to 3.3% in the United States. Issue-based political communication is the most prevalent form but reduces engagement by 14%, while personalized content decreases engagement by 21%. Party-political posts cannot be evaluated reliably due to small case numbers. Although political posts by celebrities generally perform worse, mixed-effects models reveal a small positive within-account effect (+5%). The strongest effect is observed for politicians: support from prominent individuals increases engagement with their posts by approximately 44%. Overall, the study demonstrates that political communication by celebrities in Germany is rare and highly context-sensitive, yet offers substantial potential when politicians strategically collaborate with well-known public figures.

Keywords: Celebrities, Social Media, Political Communication, Endorsements, Computational Social Science

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# 1 Celebrity Endorsements in the 2024 U.S. Election Campaign

Once again, the 2024 U.S. election campaign was marked by celebrities from various sectors of society publicly endorsing a political party and, in some cases, actively supporting election campaigns (Foggat, 2024). As with many other aspects of political communication, these so-called endorsements are increasingly shifting into the digital sphere, where they are disseminated in various forms via social media. Online, endorsement messages achieve substantial reach; on the other hand, their mobilizing effects can be more clearly operationalized using metrics such as likes and comments. Media attention and coverage of these different forms of support have been extensive, extending to the publicly debated hypothesis that Taylor Swift could significantly influence the election outcome (Foggat, 2024). The endorsement post of the currently most successful musician in support of the Democratic candidate Kamala Harris received 11.5 million likes. From a political science perspective, this development raises pressing questions about its implications for elections and democratic processes and calls for empirical data analysis. While the culture of political endorsements is established and well documented in the United States (Garthwaite & Moore, 2012, p. 355), considerably less research has examined the relevance of this phenomenon in Germany. Although expressions of political support by unions, business associations, and other civil society actors have traditionally played an important role, the political communication of famous individuals from across all areas of society also warrants closer examination.

Against this background, two research questions emerge: first, how established and empirically observable is political support by celebrities on social media in Germany compared to the United States? Second, how does the involvement of celebrities and other public figures in political communication on social media affect the performance of these posts?

# 2 Celebrities as Political Actors on Social Media: Conceptual Framework and State of Research

In order to examine the research questions outlined in the introduction, a theoretical framework is required that draws on several strands of existing research on political endorsements on social media. To this end, the following section systematically outlines the key dimensions of the phenomenon and reviews the current state of scholarly knowledge associated with each of them.

## 2.1 Digital Opinion Leadership: Conceptual Foundations

Fundamental explanations of why certain individuals can influence the opinions and actions of others are provided by the concept of opinion leadership. This concept is applied in numerous sociological and communication science models and posits that specific individuals (opinion leaders) exert a disproportionate influence on the attitudes and decisions of others. They act as intermediaries by receiving information from the media or other sources, evaluating it, and passing it on within their social environment. Their influence is typically based on expertise, credibility, or social connectedness rather than on formal authority (Duckwitz, 2023, pp. 60–61; Lazarsfeld et al., 1949).

Building on this foundational concept, and in combination with the ongoing digitalization of society, the notion of digital opinion leadership has emerged. Digital opinion leaders influence public and private opinions, including in the political domain, and primarily exert this influence through digital channels (Duckwitz, 2023, pp. 60–61; Garthwaite & Moore, 2012, p. 358; Moritz, 2024, pp. 11–12; Arnesson & Reinikainen, 2024, p. 2). In summary, sociology and communication studies assume that prominent individuals can leverage their popularity to promote products, opinions, and worldviews, as well as political parties, issues, and candidates (Friedrich & Nitsch, 2019, p. 4875; Duckwitz, 2023, pp. 60–61; Garthwaite & Moore, 2012, p. 358).

Existing empirical research on politics and celebrity also draws on the concept of opinion leadership and further refines it with regard to political communication (Friedrich & Nitsch, 2019; Friedrich & Steinleitner, 2015; Schmuck et al., 2022). Within the context of endorsements, scholars differentiate between campaigns aimed at increasing voter turnout in general, support for political candidates, parties, or issues, and politicians who themselves are considered celebrities (Friedrich & Nitsch, 2019, p. 487; Duckwitz, 2023, pp. 60–61; Wheeler, 2018, pp. 5–ff.). This analysis follows this heuristic and, in line with the two research questions, examines support for politicians, parties, or political issues on social media by famous individuals. The communicative potential of these actors can be theoretically grounded in the concept of digital opinion leadership.

## 2.2 Country-Specific Research on Political Endorsements on Social Media

The basic concepts of the existing body of research on endorsements and opinion leadership discussed above constitute the theoretical foundation of this analysis. Beyond this,

however, the question arises to what extent digital opinion leadership has been examined across different countries and political cultures.

As outlined in the introduction, the phenomenon of celebrity endorsements is far more deeply rooted in the United States (Garthwaite & Moore, 2012, p. 355). Accordingly, it is not surprising that many studies have examined the influence of prominent individuals on U.S. elections (e.g., Oprah Winfrey's endorsement of Obama in 2008 or Taylor Swift in 2024). These studies report, in part, significant effects on vote shares, campaign donations, and voter turnout (Garthwaite & Moore, 2012; Pew Research Center, 2007; Fung, 2024). The focus of this research has been predominantly on presidential elections. More generally, qualitative interviews and case studies conducted in the U.S. context suggest that many celebrities use their public visibility to promote political education, voter registration, and political mobilization. However, empirical validation using voter turnout as the dependent variable remains a research gap (Spillane, 2024).

At the same time, normative arguments can be found that fundamentally question the relevance of celebrity endorsements for electoral outcomes. In addition to the positive effects outlined above, it is assumed that celebrity support may also have negative consequences—for instance, in the case of Hillary Clinton's 2016 campaign, where her close association with numerous celebrities was perceived as elitist (Foggat, 2024; Garthwaite & Moore, 2012). By contrast, studies conducted in the German context report little to no effect of celebrity voting recommendations (Friedrich & Nitsch, 2019; Duckwitz, 2023), while influencers tend to address broader societal issues such as climate protection or gender equality rather than electoral politics. Comparative European studies likewise find only limited effects of celebrity appeals on voting intentions (Friedrich & Steinleitner, 2015; Schoen & Faas, 2009). Research in non-Western contexts, such as Taiwan (Henneberg & Chen, 2008) and South Africa (Madinga et al., 2021), highlights cultural differences: in these cases, celebrity endorsements are considered a central marketing instrument and positively affect voting intentions primarily through perceived credibility.

The strong focus of the existing literature on the United States leads to the first hypothesis of this study, which seeks to compare differences in political communication on social media by opinion leaders in Germany and the United States.

## 2.3 Social Media and Political Endorsements

Endorsements and the political participation of prominent individuals are not new phenomena. However, the existing literature attributes significantly increased impact to them due to the affordances and reach of social media (Aw & Labrecque, 2020; Spillane, 2024). Accordingly, more recent analytical approaches focus explicitly on social media as the primary object of investigation, whereas classical endorsement studies often operate without a platform-specific perspective. In addition to Facebook, influencers primarily use Instagram, TikTok, and YouTube to disseminate political content (Duckwitz, 2023; Schmuck et al., 2022). These platforms therefore constitute a central focus of research. Globally, Instagram ranks just behind Facebook and YouTube, with more than 55% of internet users aged 16 and older using the image-centered platform at least once per month (We Are Social & Meltwater, 2025).

Detailed empirical findings indicate that mechanisms such as interactive appeals on social media can increase voting intentions (Pöyry & Reinikainen, 2024), while the simplification of political content may simultaneously foster political cynicism and political interest (Schmuck et al., 2022). Discourse-analytical studies reveal ambivalent perceptions of political influencers, situated between their potential to democratize political communication and concerns about the erosion of journalistic standards (Arnesson, 2024). In sum, both political and non-political endorsements in the digital age predominantly take place on social media platforms, where they are broadly communicated and disseminated. This observation underpins the focus of the present study on social media in general and on Instagram in particular.

## 2.4 Methodological Perspectives on Political Endorsements

The political communication of famous individuals is examined using a wide range of methodological approaches. Experiments dominate the measurement of endorsement effects (Friedrich & Nitsch, 2019; Pöyry & Reinikainen, 2024) and typically find no or only very weak effects. Surveys (Fung, 2024; Pew Research Center, 2007; Madinga, 2021), econometric analyses (Garthwaite & Moore, 2012), and qualitative approaches such as case studies (Duckwitz, 2023; Henneberg & Chen, 2008), discourse analyses (Arnesson, 2024), and focus groups (Nisbett & DeWalt, 2016) provide insights into situational perceptions and identify communicative strategies. Theoretical contributions (Marsh et al., 2026; Wheeler, 2018) offer conceptual frameworks and constitute an important basis for

interpreting empirical findings, including those of the present study. In this context, typologies of different forms of celebrity politics are particularly prominent.

Overall, research on endorsements and celebrity support spans multiple academic disciplines and is primarily situated within communication and media studies. While many studies rely either on experimental designs or qualitative approaches with small sample sizes, large-scale empirical data analyses capable of quantifying the phenomenon remain scarce. The present analysis seeks to address this research gap.

## 2.5 Research Question and Hypotheses

Based on the state of research outlined above and the underlying conceptualization of digital opinion leaders in political communication, the following research objectives and hypothesis structure are derived. As existing research has focused primarily on the potential effects of endorsements, the present analysis adopts a more fundamental approach. It aims to measure the extent to which political endorsements take place on social media in Germany and to assess whether this phenomenon is comparable to the situation in the United States in terms of both quantity and quality. In addition, the analysis examines whether so-called opinion leaders achieve different levels of performance with political and non-political posts, and whether politicians attain higher performance when their posts visibly reflect support from celebrities.

From these considerations, four hypotheses are formulated:

1. The frequency with which celebrities publish political content on social media differs between Germany and the United States.
2. In Germany, different forms of political communication by celebrities - issue-based, party-political, or personalized - vary in their relevance and are associated with different levels of performance.
3. The performance of posts by opinion leaders differs depending on whether the content is political or non-political.
4. The performance of posts by politicians differs depending on whether the posts visibly indicate support from prominent individuals.

# 3 Methodology and Data Basis

## 3.1 Data Basis

Based on the hypothesis structure outlined above, the data foundation of this analysis is divided into two parts. One sample consists of Instagram posts by prominent individuals from Germany and the United States, while the second sample comprises Instagram posts by politicians during the campaign for the 2025 German federal election. As both samples were collected on Instagram using the same procedure, the technical data collection process is described first, followed by a detailed presentation of the resulting datasets.

The selection of Instagram as the empirical basis of this study is motivated by the platform's high relevance and extensive user base. In the third quarter of 2025, the Meta-owned platform ranked second worldwide in terms of popularity among social media platforms, with a market share of 16.4%, trailing only WhatsApp. Three billion users access Instagram at least once per month; only YouTube and Facebook report higher monthly usage figures (We Are Social & Meltwater, 2025, pp. 417, 419). This trend is also evident in Germany, where 40% of the population used Instagram at least weekly in 2025. As a result, Instagram became the most frequently used social network in Germany, representing an increase of three percentage points compared to the previous year (ARD/ZDF Media Commission, 2025).

Posts from both subsamples were collected from Instagram using the Zeeschuimer add-on and the 4CAT software and subsequently prepared for analysis (Peeters, 2023; Peeters et al., 2023). For each post, the caption and additional metadata - such as interaction metrics, hashtags, and the time of publication - were stored. One technical feature of Instagram that affects the data basis concerns so-called Reels, short video formats that users can publish. These are only displayed on a profile's main overview page if users have not disabled this option (Instagram, 2024). As only posts visible on the profile overview page could be extracted for technical reasons, Reels are included in the dataset only if they also appear as previews in the main profile grid. In addition, since 2021, so-called collaboration posts have become a common feature on Instagram, allowing users to publish posts jointly with other accounts (Sides, 2023). This also affects the data structure, as socially, professionally, or politically connected individuals may appear together in a single post. As a result, some posts are included multiple times in the dataset. However, since these posts are distributed to all followers of the participating accounts and thus generate independent reach and communicative potential, treating them as separate observations is

appropriate considering the research objectives. The following section provides a detailed description of the characteristics of both subsamples.

### 3.1.1 Dataset I: Instagram Posts by Celebrities from the United States and Germany

For the first two hypotheses, the empirical basis of this study consists of 169,333 Instagram posts by 131 German celebrities and 161,229 posts by 69 U.S. celebrities. In both countries, personal accounts were identified and ranked in descending order according to their number of followers, with citizenship serving as the key selection criterion. As no comprehensive and verified list of individuals with the highest numbers of Instagram followers exists for either country, the sample was constructed using a combination of existing rankings and additional manual research.

For the German sample, all individuals holding German citizenship and having more than one million followers were included. A similar procedure was applied to the comparative U.S. sample. However, due to the larger population size of the United States and the global reach of U.S. popular culture, American celebrities and influencers typically have substantially higher follower counts. Consequently, all German profiles were collected first, after which accounts of individuals with U.S. citizenship were added until a comparable sample size was achieved. Overall, the dataset covers posts published between June 6, 2011, and January 23, 2025.

### 3.1.2 Dataset II: Instagram Posts by Politicians during the 2025 German Federal Election Campaign

For the political component of this study, a dataset comprising 149,465 Instagram posts by 2,618 German politicians and 270 political parties was collected using the data collection procedure described above. The dataset covers the campaign period for the 2025 German federal election from November 6, 2024, to February 23, 2025. To construct the dataset, an attempt was made to identify Instagram profiles for all 4,506 candidates officially reported by the Federal Returning Officer. This was successful in 58.1% of cases.

In addition, the official federal party accounts as well as the profiles of the individual state party branches, if available, were included for all parties contesting the election. The dataset was further enriched with additional information on candidates drawn from the Federal Returning Officer's database for the election to the 21st German Bundestag.

To ensure data quality, a stratified sample evaluation was conducted. In this process, 6,000 posts were manually reviewed, focusing on the accuracy of post metadata and the

identification of potential structural or technical errors. Based on this evaluation sample, both incorrect metadata and structural or technical errors could be ruled out.

## 3.2 Data Classification and Analysis

After completing data collection and processing, the post texts and hashtags of the extracted posts were subjected to a prompt-based[1] classification in order to examine (a) the involvement of prominent individuals and (b) the presence of potentially political content. For this purpose, the model GPT-4o mini was used via the OpenAI API. The quality of the automated classifications for both datasets was assessed through cross-validation against human coding, which served as the ground truth.

To this end, posts were selected from both datasets using a stratified sampling procedure. The sample was drawn so that the distribution of accounts and post formats (video vs. photo) matched the distribution in the full dataset. Agreement between automated classification and human coding was evaluated using k-fold cross-validation (k = 5) to assess the model's generalizability and robustness. Splitting the data into five equally sized folds[2] and running five training and test iterations provided a robust estimate of model performance. This approach helps ensure that results are not driven by a particular train–test split but instead reflect the model's ability to generalize to new data (Wong & Yeh, 2020, p. 1586).

Across folds, different subsets of posts were provided as training examples, while the classifier was evaluated on the remaining posts in the stratified sample. To quantify agreement between the model's classifications and human decisions, accuracy, precision, recall, and the F1 score were calculated. For each metric, the mean value across the five folds was reported. For the final classification of the full dataset, the classifier was again provided with human-labeled posts as reference data in order to optimize classification performance.

While the technical approach to classification and evaluation was identical for both samples, the specific classification schemes differ. Accordingly, the evaluation results are discussed separately for each dataset.

[1] The prompt used for classification is included in the code (see Appendix A).

[2] The training and test datasets comprised the following cases: for the U.S. sample, 1,896 training cases and 7,584 test cases; for the German sample, 1,896 training cases and 7,853 test cases; and for the election campaign sample, 1,963 training cases and 3,810 test cases.

### 3.2.1 Classification of Dataset I: Instagram Posts by Celebrities from the United States and Germany

Posts by celebrities from the United States and Germany were classified according to whether they contain political communication. To do so, the distinction identified in the literature as established—support for political actors, political issues, and political parties—was operationalized using a prompt. A multi-class approach was applied, allowing individual posts to be assigned to more than one category. The content structure of the classification prompt was iteratively refined and evaluated in order to optimize classification outcomes.

The conceptual definition of political content was deliberately broad. This includes, for example, posts that primarily call for donations to charitable causes, as such communication can have societal effects and cannot be fully separated from a political dimension. This inclusive classification strategy follows the guiding principle of avoiding the under-detection of political communication.

To evaluate the first dataset, 9,470 posts were selected from the German dataset using stratified sampling and were manually coded to provide a ground-truth benchmark for assessing classification performance. The cross-validation results are promising: all standard evaluation metrics indicate very strong performance. Across the full sample, there were few false-positive classifications, while more than 99% of the true positive cases were correctly identified (see Fig. 1). A similar pattern emerges at the level of individual categories: nearly all posts belonging to one of the three classes were detected, and false negatives are extremely rare (see Fig. 2).

However, precision drops notably in the categories "party" and "person," as the classifier repeatedly produces false positives in these classes. The overall magnitude of posts incorrectly classified as political is in the single-digit percentage range and thus only slightly above the estimated share of genuinely political posts. Given the very high recall values, it can be assumed that hardly any truly political cases were missed. At the same time, only a limited number of posts were classified as false positives (n = 2,113). Therefore, the output of the automated classification was reviewed again, and all false-positive cases were manually corrected. This ensured a high-quality data basis for the subsequent analysis.

As a result of this manual cleaning of false-positive assignments, the F1 score increased substantially across all categories. In particular, in the previously error-prone classes

“person” and “party,” the F1 score reached 1.0, while in the “issue/content” category it increased to approximately 0.92. This review was not conducted to adjust the model post hoc, but to ensure the most precise and reliable data basis possible for the analyses that follow.

| | |
|---|---|
| Accuracy | 0.9943 |
| Precision | 0.9963 |
| Recall | 0.9943 |
| F1-Score | 0.9951 |

*Figure 1: Evaluation metrics of the automated classification using k-fold cross-validation (k = 5), weighted averages, authors’ own calculation, n = 5 × 7,583 = 37,915.*

| **Category** | **Accuracy** | **Precision** | **Recall** | **F1-Score** |
|---|---|---|---|---|
| Person | 0.9893 | 0.0368 | 1.0000 | 0.0708 |
| Issue | 0.9944 | 0.8232 | 0.8439 | 0.8333 |
| Party | 0.9943 | 0.2214 | 1.0000 | 0.3556 |

*Figure 2: Evaluation metrics of the automated classification by category using k-fold cross-validation (k = 5), authors’ own calculation, n = 5 × 7,583 = 37,915.*

The evaluation procedure described above was applied analogously to the U.S. sample in a second step. The structure of the prompt, the multi-label setup, and the iterative fine-tuning of the classification scheme were fully identical to the approach used for the German dataset. The evaluation metrics likewise indicate very strong performance; only minor deviations were observed, primarily in the magnitude of individual precision values.

As in the German case, false-positive classifications in the U.S. sample were manually corrected, resulting in a substantial increase in the F1 scores across all categories. Consequently, a consistent and methodologically validated data basis is available for both countries. Detailed evaluation results for the classification of the U.S. sample are provided in Appendix F.

### 3.2.2 Classification of Dataset II: Instagram Posts by Politicians during the 2025 German Federal Election Campaign

| | |
|---|---|
| Accuracy | 0.9876 |
| Precision | 0.0917 |

| | |
|---|---|
| Recall | 0.7271 |
| F1-Score | 0.1622 |

*Figure 3: Evaluation metrics of the automated classification using k-fold cross-validation (k = 5), weighted averages, authors' own calculation, n = 5 × 3,807 = 19,035.*

Analogous to the first sample of Instagram posts by celebrities, the dataset of political posts from the most recent German federal election campaign was also subjected to classification. Using the GPT model, posts were classified according to whether they visibly indicate support by a well-known public figure. This procedure is grounded in the concept of digital opinion leadership outlined above. The content structure of the classification prompt[3] was iteratively refined and evaluated in order to further optimize classification performance.

As with the first dataset, a stratified sample was created for the evaluation of the second dataset and manually coded to again assess model performance against a ground-truth benchmark. In total, 3,807 posts were evaluated. The results of the cross-validation procedure are generally satisfactory; however, once again, manual correction of false-positive cases proved to add substantial value. While accuracy is very high and recall is at an acceptable level, precision declines markedly due to a considerable number of false-positive classifications.

Here, too, the share of false-positive posts lies in the single-digit percentage range (1.06%) and is therefore approximately one percentage point higher than the estimated proportion of truly relevant posts in which support by opinion leaders is visible (0.1%). Given the manageable number of false-positive cases (n = 1,590) and the acceptable recall values, manual post-processing was feasible. This step was taken to improve the overall quality of the classification and to enable valid hypothesis testing and effect estimation. Through this manual reclassification, all false-positive cases were removed. For the central F1 score - which is sensitive to both false positives (precision) and false negatives (recall) - this refinement resulted in an increase from 0.1622 to 0.84.

## 3.3 Applied Analytical Methods

To test the hypotheses, the resulting datasets - including the newly created classes serving as independent variables - were examined using descriptive and inferential statistics. The

[3] The prompt used for classification is included in the code (see Appendix A).

dependent variable was the interaction rate[4], defined as the ratio of a post's likes and comments to the follower count of the corresponding account. Interaction rate was used as a measure of performance in order to avoid biases arising from differences in follower counts across accounts and the resulting variation in absolute interaction numbers (Gräve, 2019, p. 4). To address non-normality and to reduce the influence of individual posts with very high engagement rates, the engagement rate was additionally log-transformed (West, 2022).

# 4 Political Communication by Opinion Leaders: Analysis and Results

Based on the methodology outlined above, the dataset is analyzed with regard to the stated hypotheses. The structure of the analysis follows the sequential examination of these research questions, beginning with a country-specific comparison of frequencies (H1), followed by an analysis of differences across issue-based, party-political, and person-centered content (H2), and concluding with an assessment of the performance-related hypotheses concerning celebrities and political communication (H3 and H4).

## 4.1 Frequency Analysis of Political Communication by Opinion Leaders in Germany and the United States

| USA | | | | | | |
|---|---|---|---|---|---|---|
| Category | Classification (final) | | Classification (raw) | | Corrected | |
| | Absolute | Relative | Absolute | Relative | Absolute | Relative |
| Issue | 5.058 | 3,14% | 5.371 | 3,33% | 313 | 0,19% |
| Person | 639 | 0,40% | 2.031 | 1,26% | 1.394 | 0,86% |
| Party | 43 | 0,03% | 120 | 0,07% | 77 | 0,05% |
| Political communication (total) | 5.332 | 3,31% | 6.555 | 4,07% | | |
| **Germany** | | | | | | |
| Category | Classification (final) | | Classification (raw) | | Corrected | |
| | Absolute | Relative | Absolute | Relative | Absolute | Relative |
| Issue | 2.813 | 1,66% | 3.268 | 1,93% | 455 | 0,27% |
| Person | 181 | 0,11% | 1.614 | 0,95% | 1.425 | 0,84% |
| Party | 26 | 0,02% | 250 | 0,15% | 224 | 0,13% |
| Political communication (total) | 2.920 | 1,72% | 4.807 | 2,84% | | |

*Figure 4: Frequency table of Instagram posts by celebrities containing political communication, differentiated by issue-based, party-political, and person-centered categories; multiple classifications per post were possible. "Classification (raw)" refers to the results of the initial automated classification prior to manual validation, "Corrected" indicates the number of posts adjusted*

[4] Interaction rate = (Likes + Comments) / Number of followers (Gräve, 2019, p. 4)

*during the subsequent validation process, and "Political communication (total)" represents the number of posts for which at least one of the three categories was assigned. Germany: n = 169,333; United States: n = 161,229.*

The first hypothesis examines whether the frequency of political posts by prominent individuals on social media differs between Germany and the United States. As shown in Figure 4, nearly twice as many posts were identified as political in the U.S. sample compared to the German sample. This pattern is also reflected in the relative shares: 3.31% of U.S. posts contain political content, whereas only 1.72% of posts in the German sample do so.

Hypothesis 1 can therefore be clearly supported. A substantial difference exists between the two countries with regard to the frequency of political communication by prominent actors. This finding is consistent with the literature, which suggests that political engagement by celebrities is more deeply embedded in U.S. political culture and is consequently more visible in their social media presence. The high quality of the automated classification and the subsequent manual validation further support the methodological robustness of the results, allowing the identified differences to be considered reliable.

When interpreting these cross-national differences, it must be taken into account that the samples were not drawn randomly, due to the absence of official and comprehensive lists of high-reach social media accounts. Profile identification relied on rankings and manual research and may therefore be incomplete. Despite the large number of observations, this limitation constrains the generalizability of the comparison and should be considered accordingly.

At the same time, the analysis shows that political communication by celebrities is relatively rare in both countries. The vast majority of published content is non-political, with political posts accounting for only a low single-digit percentage overall. This observation is central for the interpretation of the subsequent hypotheses: while the occurrence of political communication provides a valid empirical basis for analysis, its potential societal impact must be understood in the context of a much larger volume of non-political content. This structural marginality of political posts should therefore be consistently taken into account in the following analysis steps.

## 4.2 Party, Person or Issue?

The second hypothesis examines whether different forms of political communication by opinion leaders in Germany—namely the presentation of political issues, party-political

communication, or person-centered political communication—play distinct roles and are associated with different levels of performance. The first part of the hypothesis is examined using an exploratory descriptive approach, while the performance analysis is conducted through inferential statistical testing based on the descriptively derived indicators.

### 4.2.1 Descriptive Analysis

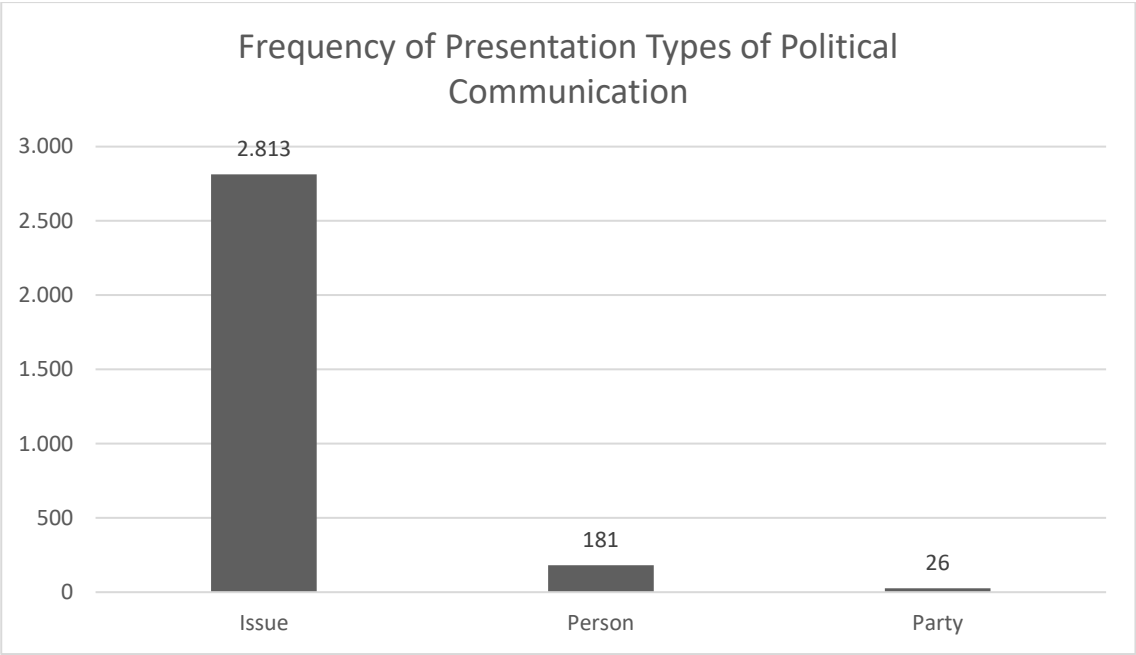


*Figure 5: Frequency of presentation types of political communication by opinion leaders in Germany on Instagram, n = 169,333.*

The frequency analysis of the classification shows that, in Germany, political communication related to substantive issues predominates (n = 2,813), whereas party-political (n = 26) or person-centered posts (n = 181) play a much smaller role (see Fig. 5). With regard to the first part of Hypothesis 2, it can therefore be concluded that the three categories indeed differ in their relative importance, with issue-based communication being the only clearly relevant type in substantive terms. A very similar pattern emerges for the United States, where issue-based statements also dominate.

The descriptive results initially support the assumption that different forms of political communication in Germany are associated with different levels of performance. Issue-based political posts achieve the highest interaction values overall, with a mean of M = 0.0141 and a median of Md = 0.0081. Person-centered political posts perform somewhat worse (M = 0.0132, Md = 0.0063) and exhibit higher volatility. Party-political content shows the highest mean (M = 0.0258) and the highest median (Md = 0.0101), but also the greatest dispersion, as shown by the high variance in Interaction Rates (Figure 6, SD =

0.0599). Crucially, party-political posts are extremely rare in the dataset (n = 26; 0.02%). Due to this small number of cases, means, medians, and dispersion measures are highly sensitive to outliers and therefore only of limited reliability. The distribution of (log-transformed) Interaction Rates by category is visualized in Fig. 6. The generalizability of these descriptive findings is assessed next using inferential statistical tests.

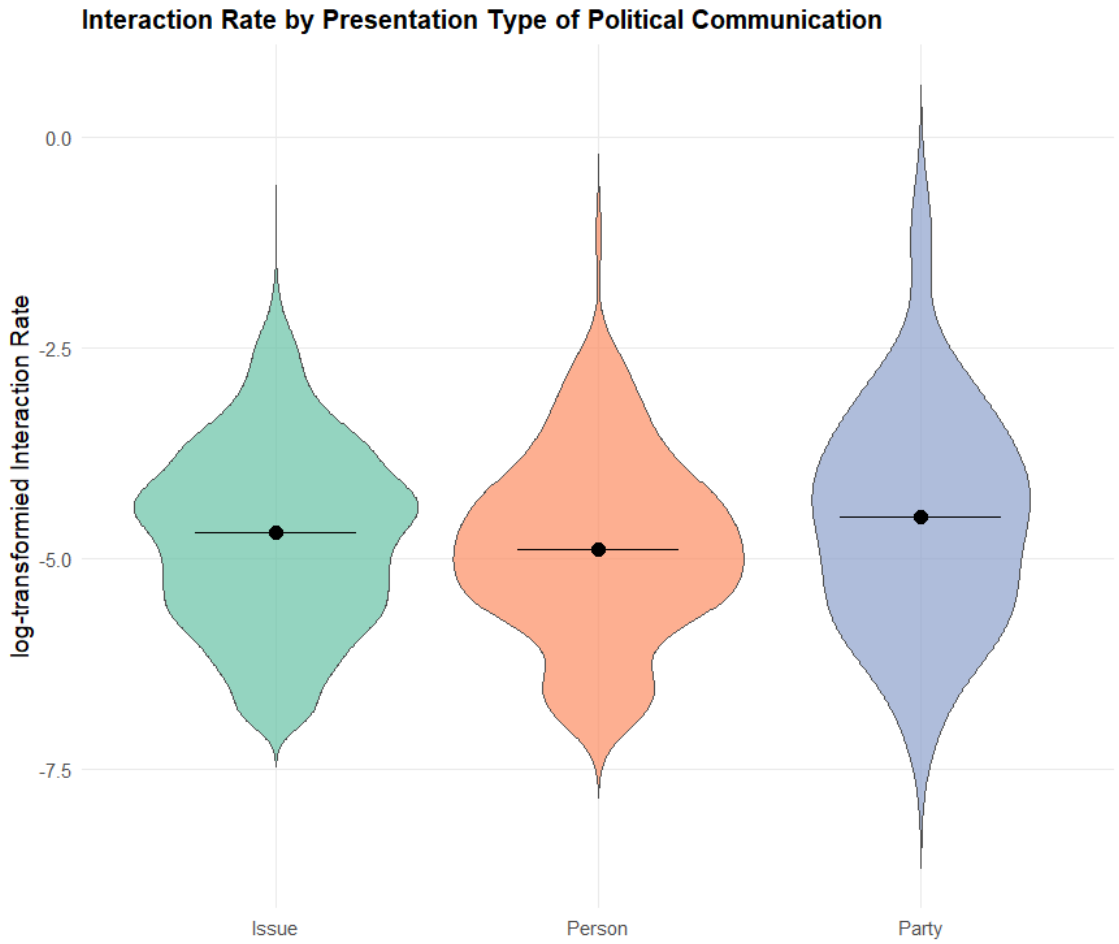


*Figure 6: Log-transformed Interaction Rates by presentation type of political communication, n = 3,020.*

### 4.2.2 Inferential Analysis

To test the hypothesis, a linear regression model was estimated with the log-transformed Interaction Rate (log(rate + 0.001)) as the dependent variable. The independent variables consist of three dichotomous indicators capturing the presentation type of political communication: issue-based, person-centered, and party-political content. The reference category comprises posts that do not contain any issue-based, person-centered, or party-political labeling.

As part of the regression diagnostics, the core OLS assumptions were assessed, including linearity of the relationship, homoskedasticity of the residuals, and independence of errors. Graphical and statistical tests indicated no substantial violations of linearity. However, tests of homoskedasticity (e.g., residual plots) revealed pronounced heteroskedasticity. In addition, the Durbin–Watson test indicated strong positive autocorrelation of the residuals (DW = 0.32, $p < 0.001$). To address these issues, heteroskedasticity- and autocorrelation-consistent standard errors were computed using the Newey–West method (lag = 5) (Newey & West, 1987). The regression coefficients themselves remained unchanged; only the standard errors are estimated robustly.

The robust estimates reveal statistically significant negative effects for issue-based and person-centered political communication. Posts with an issue-based focus exhibit an approximately 14.3% lower Interaction Rate ($\beta = -0.155$, $p < 0.001$), while person-centered political posts show Interaction Rates around 20.0% lower than the reference category ($\beta = -0.223$, $p < 0.01$). For party-political content, the coefficient is positive ($\beta = 0.257$, corresponding to an increase of 29.3%) but not statistically significant ($p = 0.361$). This lack of significance is again attributable to the very small number of observations in this category (n = 23; 0.02%), which increases estimation uncertainty and susceptibility to bias. Overall, the results prove robust to corrections for heteroskedasticity and autocorrelation.

| Independent Variable | ß | SE | T | p |
|---|---|---|---|---|
| (Intercept) | -4.595 | 0.006 | -763.851 | .000*** |
| Issue | -0.155 | .024 | -6.368 | .000*** |
| Person | -0.223 | .084 | -2.653 | .007** |
| Party | .257 | .281 | 0.914 | .361 |

*Figure 7: Multivariate linear regression model; dependent variable = log-transformed Interaction Rate (likes + comments / followers). Standard errors were computed using the Newey–West method (lag = 5). β = unstandardized regression coefficient, SE = standard error, t = t-value. p < .05, p < .01, p < .001. $R^2 = 0.0004$, adjusted $R^2 = 0.0004$, n = 169,329.*

In summary, the inferential statistical results only partially confirm the descriptive findings. While both issue-based and person-centered political communication significantly reduce the Interaction Rate, no robust advantage can be identified for party-political content despite its comparatively high descriptive means. With regard to the two statistically significant categories, the results further indicate that issue-based political communication is associated with a smaller reduction in interaction than person-centered political statements.

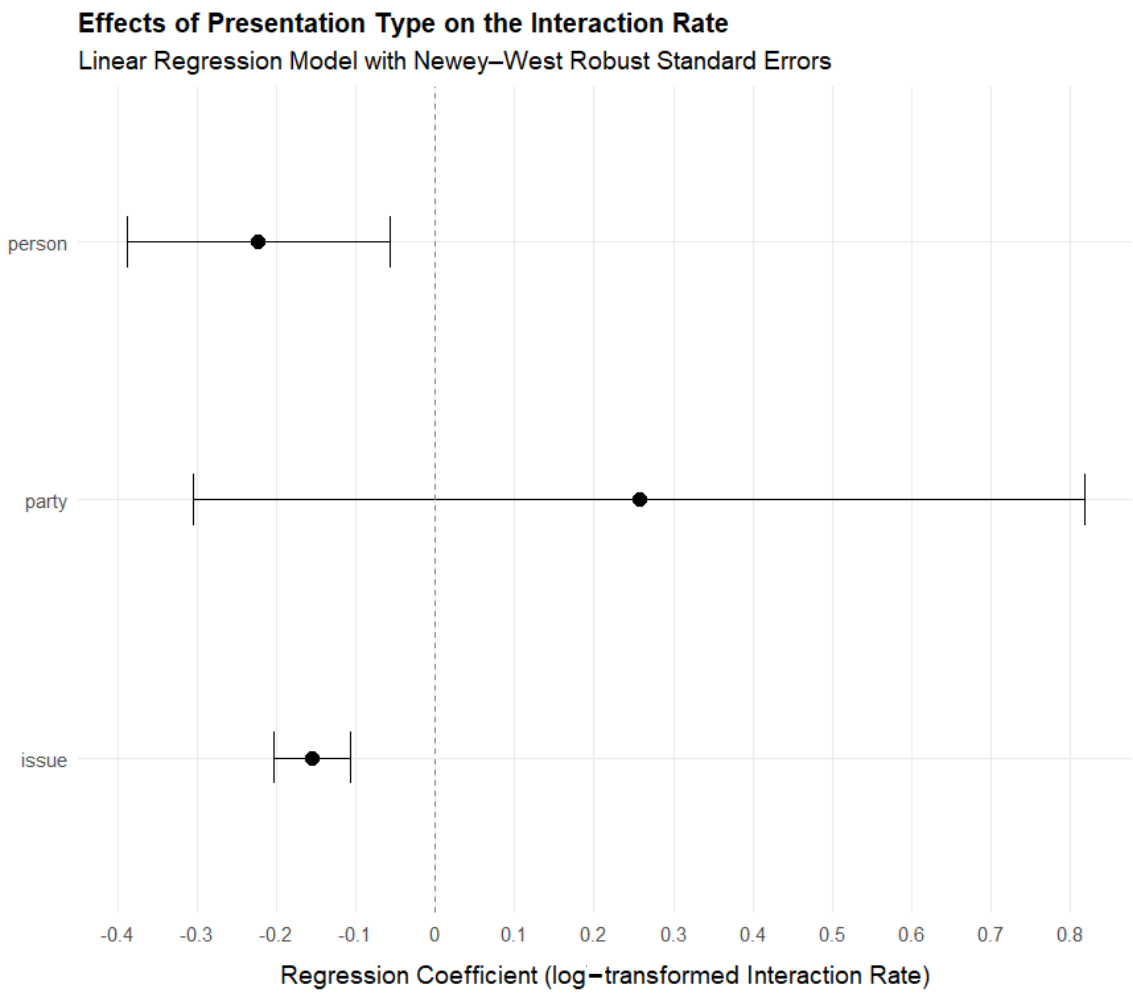


*Figure 8: Coefficients with 95% confidence intervals from a linear OLS regression with Newey–West robust standard errors (lag = 5), n = 169,329, $R^2 = 0.0004$.*

## 4.3 Political Communication: Costs or Benefits for Opinion Leaders?

The third hypothesis examines if the performance of posts by opinion leaders differs systematically depending on whether a post contains political or non-political content. To address this question, exploratory group comparisons were conducted first, followed by the estimation of a linear mixed-effects model that accounts for account-specific variance. In contrast to the previous hypothesis, the specific category of political content is not relevant in this case. The decisive factor is whether any form of political communication is present in the post.

### 4.3.1 Explorative Analysis

First, exploratory group comparisons were conducted. For this purpose, a Welch's t-test based on the log-transformed Interaction Rate was calculated. The results indicate a statistically significant difference between political and non-political posts ($t = 7.84$, $df = 3{,}025$, $p < 0.001$). Political posts exhibit a lower mean log Interaction Rate ($M = -4.75$) than non-political posts ($M = -4.59$) (see Figs. 9 and 10). This corresponds to an estimated reduction in the Interaction Rate of

approximately 17% for posts containing political content. Overall, the dataset comprises 169,333 posts from 131 accounts, of which 1.72% were classified as political.

| | n | mean | median | sd |
|---|---|---|---|---|
| non-political | 166,413 | -4.596 | -4.540 | 1.084 |
| political | 2,920 | -4.752 | -4.704 | 1.070 |

*Figure 9: Descriptive statistics (mean, median, standard deviation) of the log-transformed Interaction Rate for non-political versus political posts, n = 169,33*

However, these results should be interpreted with caution, as the assumptions of normality and independence of observations are violated. This is due to the hierarchical structure of the data, with multiple posts nested within accounts and substantial differences between accounts in terms of posting frequency and audience reach. The t-test therefore serves solely an exploratory purpose in this context.

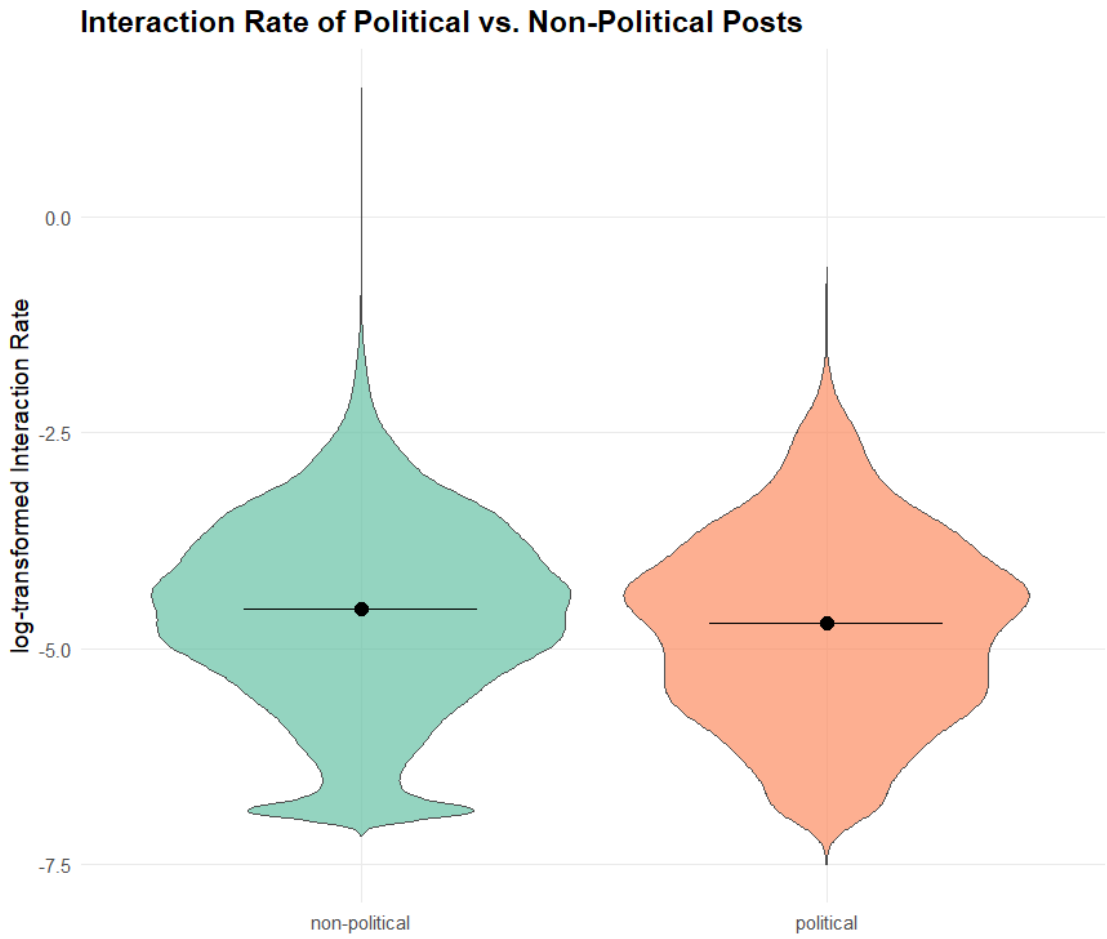


*Figure 10: Log-transformed Interaction Rates for non-political versus political posts, n = 169,333.*

### 4.3.2 Mixed-Effects-Model

To control for account-specific differences, a linear mixed-effects model with random intercepts at the account level was estimated. The dependent variable was the log-transformed Interaction

Rate, and a dichotomous indicator for political content was specified as the central fixed effect. Model diagnostics indicated an approximately normal distribution of residuals and overall acceptable homoskedasticity. The variance component of the random intercepts (Var = 0.65) suggests that a substantial share of the total variance can be attributed to differences between accounts.

In contrast to the exploratory comparison, the mixed-effects model reveals a positive and statistically significant effect of political content: within the same account, political posts achieve slightly higher Interaction Rates (β = 0.051, p = 0.0018). This corresponds to an estimated increase in the Interaction Rate of approximately 5%.

| Independent Variable | ß | SE | T | p |
|---|---|---|---|---|
| (Intercept) | -4.227 | .071 | -59.80 | <.001*** |
| Political Content | .051 | .016 | 3.121 | .001** |

*Figure 11: Linear mixed-effects model with random intercepts at the account level (n = 131). Dependent variable: log-transformed Interaction Rate. Variance components: intercept (account) = 0.651, residual = 0.746. β = unstandardized regression coefficient, SE = standard error, t = t-value. p < .05, p < .01, p < .001. n = 169,333. REML criterion = 431,655.1.*

### 4.3.3 Comparison of Analytical Approaches: Between- and Within-Account Effects

The divergent results can be explained by a conflation of between-account and within-account effects. The t-test aggregates across all accounts and is therefore biased by the fact that some accounts predominantly post political content while simultaneously exhibiting lower average levels of interaction. In contrast, the mixed-effects model controls for these account-specific baselines and isolates the effect of political content within accounts. The negative association observed in the t-test is thus primarily driven by differences between accounts rather than by a content-specific effect of political posts.

Accordingly, the hypothesis can only be partially supported. At the aggregate level, political posts are associated with lower Interaction Rates. Once account-specific differences are taken into account, however, a small but robust positive effect of political content emerges. At the same time, it becomes evident that differences between accounts exert a stronger influence on performance than the substantive distinction between political and non-political content. This underscores the central importance of account structure for the dynamics of reach and interaction.

From a practical perspective, these findings suggest that political communication does not inherently harm the performance of individual posts. Within a given account, political content is even associated with slightly higher Interaction Rates. At the same time,

accounts with a high share of political posts tend to exhibit lower average levels of interaction overall.

## 4.4 Political Communication: The Impact of Celebrity Support

The fourth hypothesis posits that posts by politicians achieve higher Interaction Rates when visible support from prominent opinion leaders is present. The empirical basis for testing this hypothesis consists of the Instagram posts by politicians and political parties from the 2025 German federal election campaign described above, which were classified with regard to potential support by opinion leaders. Overall, only a very small number of posts were identified as being supported by opinion leaders (n = 143), corresponding to 0.096% of all posts (see Fig. 12).

Despite this low prevalence, the classification can be considered highly valid due to the overall strong evaluation results of the classification procedure and the manual correction of false-positive cases described above.

| | **n** | **mean** | **median** | **sd** |
|---|---|---|---|---|
| No Support | 149,222 | -3.365 | -3.389 | 1.241 |
| Support | 143 | -3.193 | -3.305 | 1.056 |

*Figure 12: Descriptive statistics (mean, median, standard deviation) of the log-transformed Interaction Rate by support from prominent opinion leaders, n = 149,365.*

Analogous to the third hypothesis, exploratory group comparisons were conducted first, followed by the estimation of a mixed-effects model that accounts for differences between accounts.

### 4.4.1 Explorative Analysis

An initial overview is provided by a Welch's t-test based on the log-transformed Interaction Rate. Posts featuring visible support from prominent individuals exhibit an approximately 0.17 higher log-transformed Interaction Rate, implying an increase of about 18%. However, the test narrowly failed to reach conventional levels of statistical significance ($p = 0.053$), and the 95% confidence interval included zero (−0.3466 to 0.0026). Thus, while the exploratory results point toward a positive effect, they are not statistically conclusive.

As with the preceding hypothesis, the interpretability of the t-test is limited, as key assumptions - normality, independence of observations, and substantial differences between accounts - are violated due to the hierarchical structure of the data. The test therefore serves exclusively for exploratory descriptive purposes.

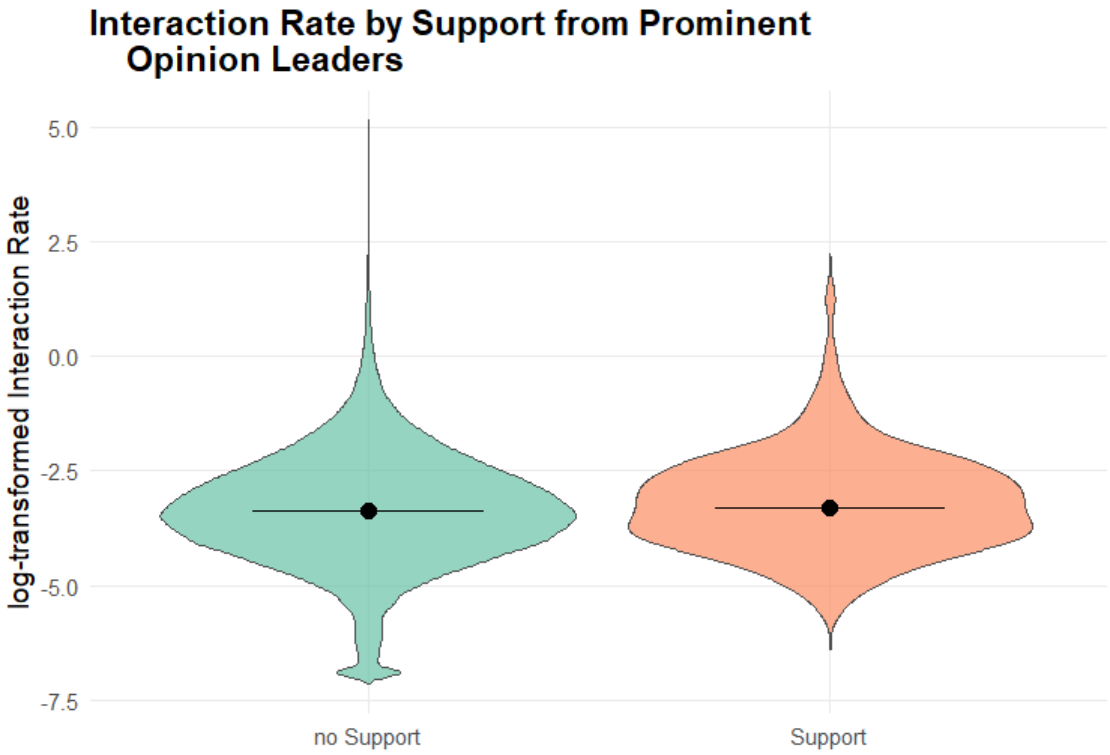


*Figure 13: Log-transformed Interaction Rates for posts with and without support from opinion leaders, n = 169,333.*

**4.4.2 Mixed-Effects-Model**

To control for account-specific differences, a linear mixed-effects model with random intercepts was estimated. The dependent variable was the log-transformed Interaction Rate, and the independent variable captured support by opinion leaders. Model diagnostics indicated overall satisfactory residual behavior; a slight degree of heteroskedasticity was observed but does not substantially affect the estimation. The intraclass correlation coefficient (ICC ≈ 0.635) indicates that approximately 63.5% of the variance in the Interaction Rate is attributable to differences between accounts. This points to a high degree of similarity among observations within the same account as well as substantial heterogeneity across accounts.

The effect of support by opinion leaders is clearly positive and highly statistically significant ($\beta = 0.367$, $p < 0.001$). This corresponds to an average increase in the Interaction Rate of approximately 44% (exp(0.367) ≈ 1.44; 95% confidence interval: +26% to +65%). The positive effect thus remains consistent and robust even after accounting for the considerable heterogeneity between accounts.

| Independent Variable | ß | SE | T | p |
|---|---|---|---|---|
| (Intercept) | -2.897 | .021 | -139.56 | <.001*** |
| Celebrity Support | .367 | .068 | 5.38 | <.001*** |

*Figure 14: Linear mixed-effects model with random intercepts at the account level (n = 2,785). Dependent variable: log-transformed Interaction Rate. Variance components: intercept (account) = 1.137, residual = 0.654. ß = unstandardized regression coefficient, SE = standard error, t = t-value. p < .05, p < .01, p < .001. n = 149,364. REML criterion = 431,655.1.*

The positive effect identified by the mixed-effects model can be illustrated using a hypothetical account with 10,000 followers. A typical post by this account achieves, on average, approximately 550 interactions in the absence of celebrity support. When support by prominent opinion leaders is visible in the post, the average number of interactions increases to roughly 800.

In interpreting these results, two aspects must be taken into consideration. First, the residuals exhibit slight heteroskedasticity. While this does not substantially compromise the estimates, an additional specification using robust standard errors or, alternatively, a negative binomial generalized linear mixed model (GLMM) would be a reasonable robustness check. Second, although the classification of support by prominent individuals is highly precise, it is not fully sensitive (recall = 72.71%).

To systematically assess robustness with respect to this classification uncertainty, a scenario-based sensitivity analysis was conducted. This analysis considered different assumptions regarding the Interaction Rates of the estimated 54 posts with undetected celebrity support. Under realistic assumptions, the effect weakens only marginally. Even under a strongly pessimistic scenario, the effect remains clearly positive. In the extreme worst-case scenario—assuming that all unidentified posts exhibit very low Interaction Rates (log-transformed values of approximately −5, corresponding to the weakest observed posts)—the coefficient is reduced to about $\beta \approx 0.30$, which still corresponds to an increase in the Interaction Rate of roughly 35%. Across all scenarios, the results therefore prove robust. Classification uncertainty thus primarily leads to a conservative underestimation of the effect size rather than calling the overall direction of the effect into question.

In sum, the results confirm the fourth hypothesis. Posts in which support from prominent opinion leaders is visible achieve significantly higher Interaction Rates than posts without such support. The effect is not only statistically robust but also substantively meaningful and practically relevant. At the same time, the analyses consistently show that structural differences between accounts exert a much stronger influence on performance than

content-level characteristics of individual posts. The account level thus emerges as the central determinant of interaction dynamics.

# 5 Synthesis, Implications, Conclusion

The results of this study demonstrate that the performance of political communication in the context of celebrities and politicians is highly context-dependent. It varies systematically across national settings, forms of content presentation, and the involvement of prominent support. Overall, the findings reveal a consistent yet differentiated pattern.

First, political communication by celebrities remains relatively rare and is strongly shaped by national context. In the United States, the share of political posts is substantially higher at approximately 3.31% than in Germany (1.72%), despite the comparable size of the underlying samples (Germany: $n = 169{,}333$; United States: $n = 161{,}229$). Political content is therefore more deeply embedded in the communicative practices of U.S. celebrities, whereas in Germany it represents a more marginal phenomenon. In both countries, political communication competes with a large volume of non-political content, highlighting its structurally limited visibility.

Differences also emerge with regard to the form of political communication. Among German celebrities, issue-based posts clearly dominate, while person-centered and especially party-political content occurs only rarely. Inferential statistical analyses show that both issue-based and person-centered political posts are associated with lower Interaction Rates, with the strongest negative effect observed for person-centered communication. Although party-political posts exhibit higher Interaction Rates in descriptive terms, their interpretation is statistically unreliable due to very small case numbers and high variance. Overall, these findings suggest that while political content is present, its performance is strongly contingent on the chosen format.

A particularly central finding concerns the comparison between political and non-political posts by celebrities. At the aggregate level, political posts initially appear to perform worse. However, once heterogeneity between accounts is taken into account, a different picture emerges: within the same account, political posts achieve slightly higher Interaction Rates. This apparent contradiction can be attributed to a conflation of between-account and within-account effects. Accounts that post political content more frequently tend to exhibit lower overall Interaction Rates, which biases the aggregated comparison.

Political content is therefore not an inherent disadvantage in terms of performance, but rather a characteristic of accounts with specific audience and reach structures.

The results regarding the role of celebrity endorsements are considerably clearer. Posts by politicians that visibly feature support from prominent opinion leaders achieve significantly higher Interaction Rates. The estimated effect is not only statistically robust but also substantial in magnitude: on average, Interaction Rates increase by approximately 44%. Despite pronounced variance between accounts, this effect remains stable, underscoring its high practical relevance. Prominent support can thus be considered an effective amplifier of political communication.

Taken together, the findings reveal a consistent pattern: political communication is rare, context-dependent, and strongly shaped by structural differences between accounts. Content-related factors primarily exert their influence within these structures, while the account level itself constitutes the most influential determinant of performance. Several strategic implications follow from these results. Political communication by celebrities should be designed with attention to format, with issue-based content appearing to be the most stable form. At the same time, political content can perform well within an established audience, even if it is associated with lower Interaction Rates in comparisons across accounts. Most clearly, the results highlight the potential of strategic collaboration: the integration of prominent support represents an effective lever for increasing the reach and interaction of political content. Party-political communication, by contrast, remains difficult to assess due to its rarity and volatility and should therefore be interpreted with caution.

Finally, several methodological limitations should be acknowledged. The use of the Interaction Rate and its log transformation address key biases arising from differences in account size and distributional skewness, but cannot fully eliminate them. Residual diagnostics and robust estimation techniques were employed to assess and safeguard key model assumptions. Limitations arise in particular from very small case numbers in some categories and from remaining classification uncertainty. Future research could address these issues by relying on larger datasets and alternative modeling approaches, such as generalized linear mixed models with appropriate count distributions. In the present analysis, only textual content was considered, while visual elements were not taken into account. As images may convey additional information, their inclusion could further

improve classification accuracy and, consequently, the validity of the results. Similarly, the quality of the ground truth could be enhanced through the use of multiple human coders.

Despite these limitations, the analysis shows that political communication in digital environments is shaped less by individual content characteristics than by structural and contextual factors, and that endorsements by prominent opinion leaders lead to higher levels of interaction. The findings thus contribute to a better understanding of the dynamics of political visibility and interaction at the intersection of celebrity, platform logic, and strategic communication.

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# 7 Appendix

## 7.1 Appendix A: Prompt Classification Dataset I

You are an advanced AI model for text classification.

Your task is to determine whether Instagram posts published by prominent individuals contain political content. The goal of this study is to examine the extent to which celebrities engage in political communication on social media and which communication strategies they employ.

You will receive the post text and hashtags of an Instagram post. Based on this information, classify the post into the categories defined below.

Posts may be assigned to none, one, or multiple categories (multi-label classification).

Person
This category applies if a politician is mentioned in the text and is positively or negatively evaluated or explicitly supported. Mentions or taggings using the @ symbol should only be classified as Person if the mentioned individual is clearly known as a politician. Mentions of non-political public figures must not be classified as Person. Always carefully verify whether the mentioned individual is in fact a politician.

Issue
This category applies if the post refers to a political or societal issue and evaluates or supports it positively or negatively. Examples include topics such as climate change, social justice, equality, democracy, human rights, migration, or public policy. General motivational quotes, lifestyle content, or personal reflections without political relevance must not be classified as Issue.

Party
This category applies if a political party is mentioned and positively or negatively evaluated or explicitly supported.

Base your classification only on the post text and hashtags provided. Do not infer political meaning beyond what is reasonably implied by the text.

If none of the categories apply, assign 0 to all categories. If a category applies, assign 1.

Return your answer exclusively in the following JSON format:

{"Person": 0 or 1, "Issue": 0 or 1, "Party": 0 or 1}

Posttext:"[posttext]"

Hashtags:"[hashtags]"

### 7.2 Appendix B: Prompt Classification Dataset II

You are an advanced AI model for classification.

Your task is to decide whether the texts from Instagram posts by politicians clearly indicate support or an electoral endorsement by a prominent individual.

By “prominent individual,” we mean people who are publicly well known, for example from the fields of entertainment, sports, culture, science, or media. Politicians themselves do not count as prominent individuals.

A mere mention of a prominent individual is not sufficient. Support is only present if the prominent individual explicitly or implicitly evaluates a politician, a political party, or a political position positively, expresses support, calls for a vote, or otherwise clearly signals endorsement.

Here are some already classified examples:
{example_text}

In addition, the following are some constructed examples that qualify as prominent support:
{kon_text}

For the classification, you will be provided with the post text and the hashtags.

Your task is to decide whether prominent support is present or not.

Please classify the following Instagram post according to the following category:

VIP: It is clear that a prominent individual expresses support for a political person, a political party, or a political issue, or provides an electoral endorsement.

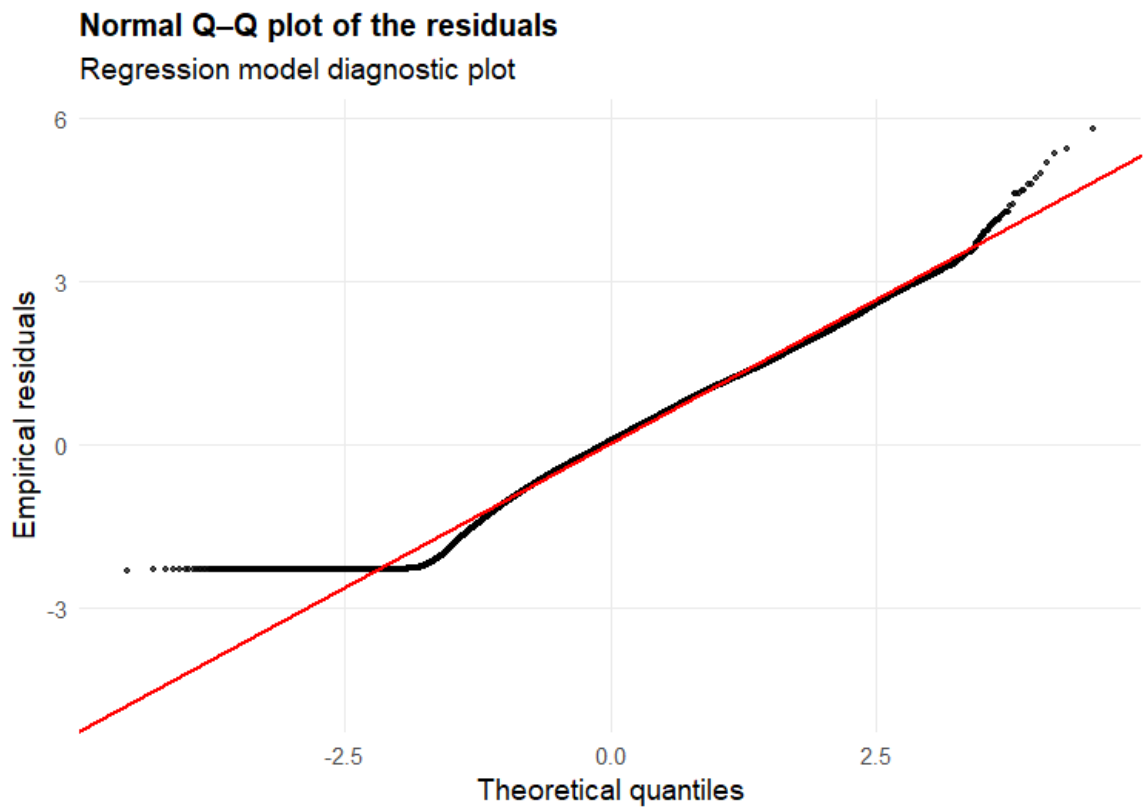
```
{
   "vip": 0 oder 1
}
```

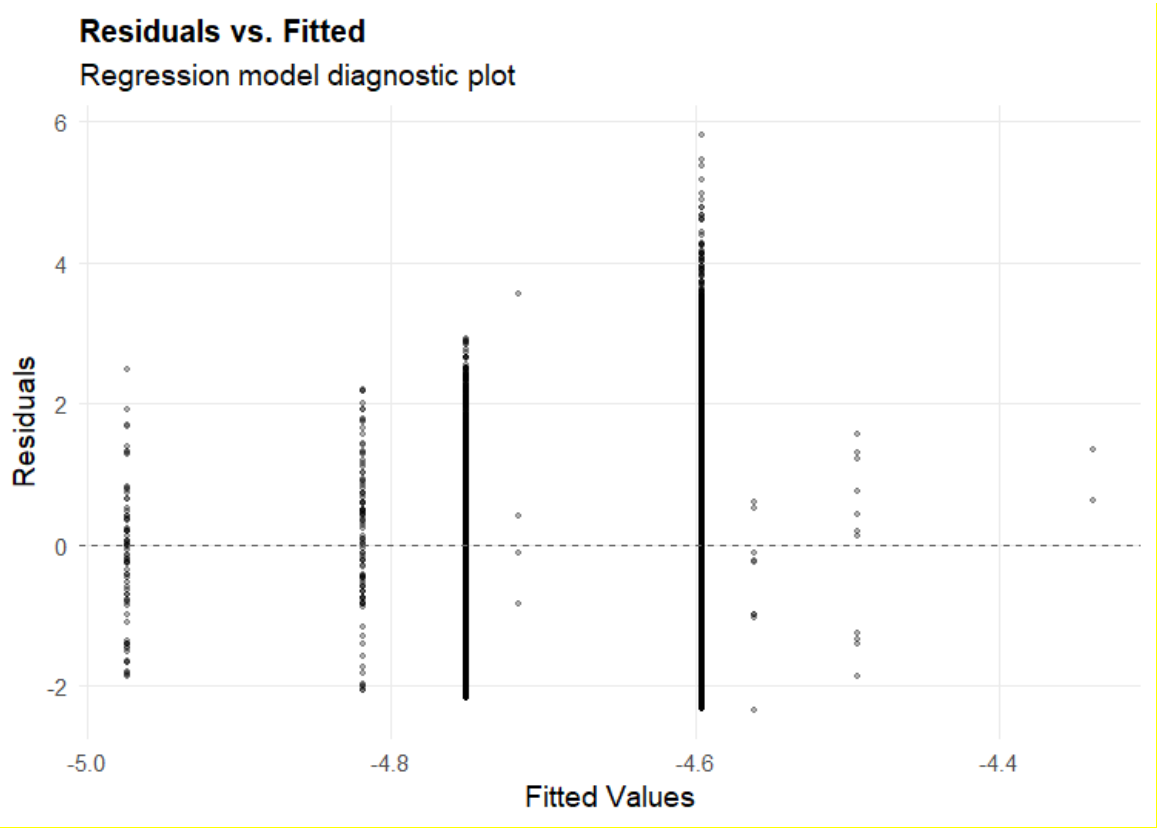
Posttext: "[body]"

Hashtags: "[hashtags]"

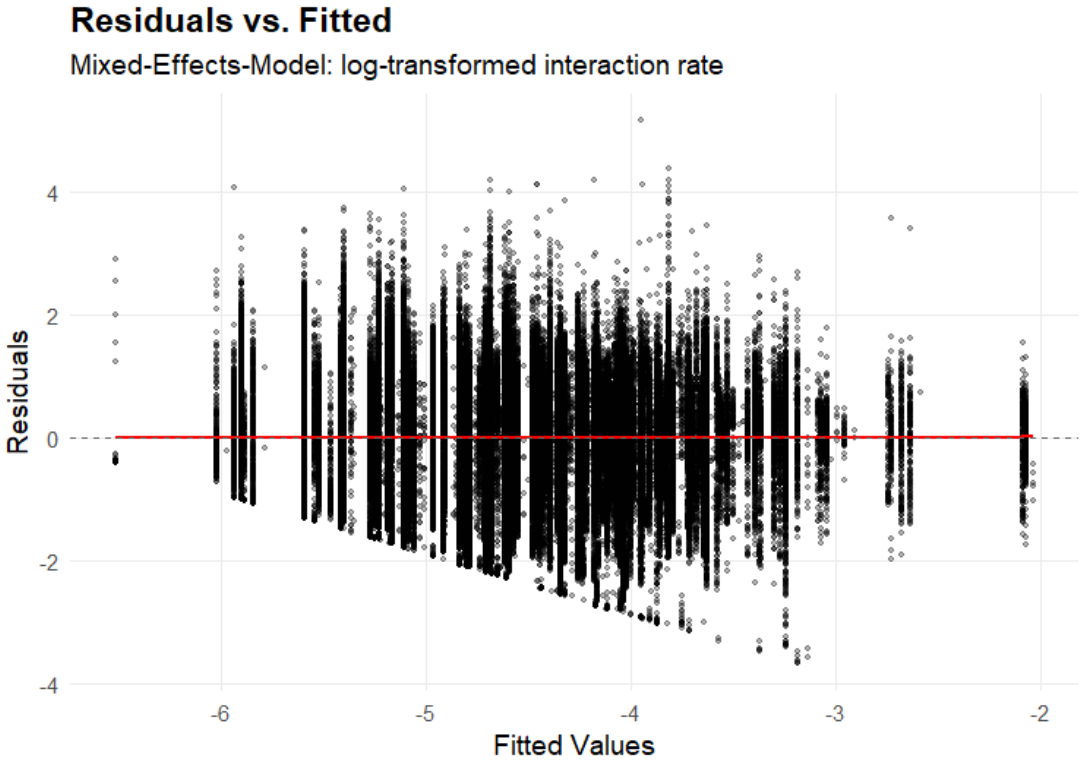
## 7.3 Appendix C: Graphical regression diagnostics

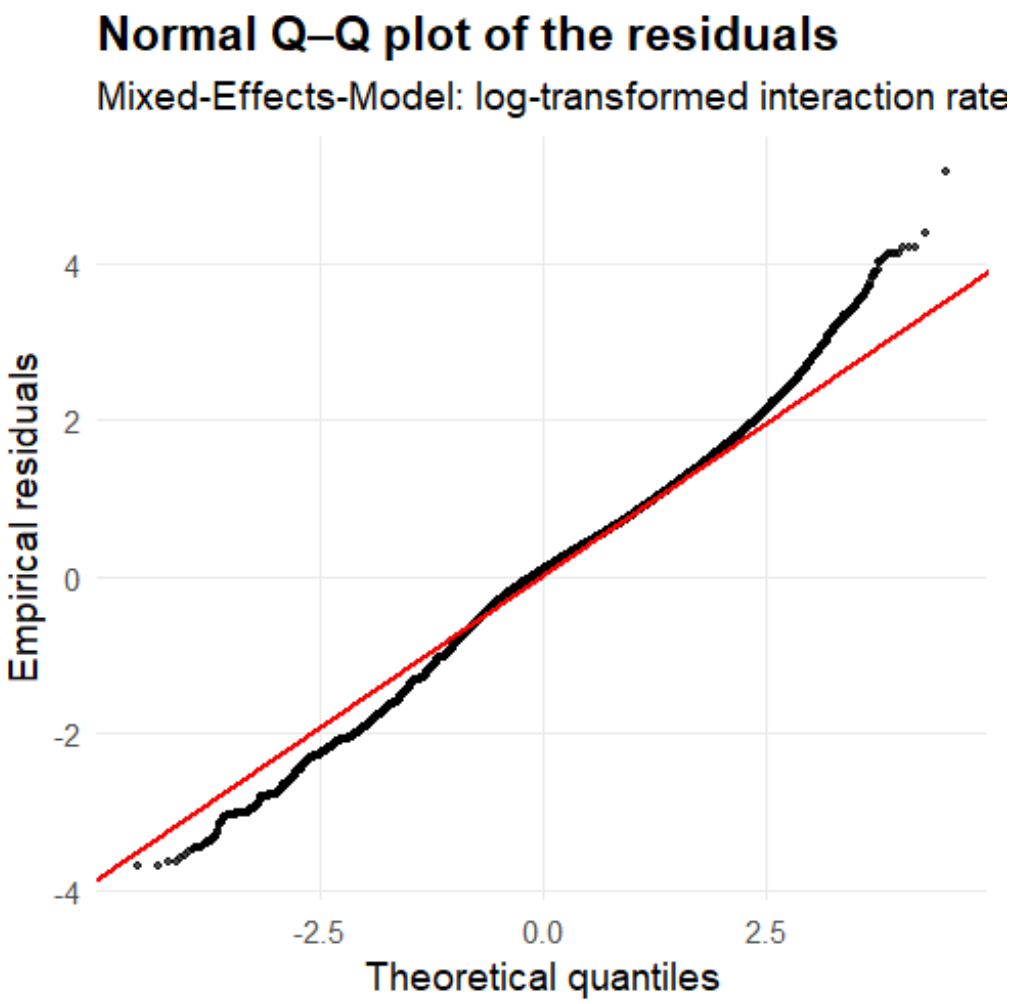
### 7.3.1 Hypothesis 2



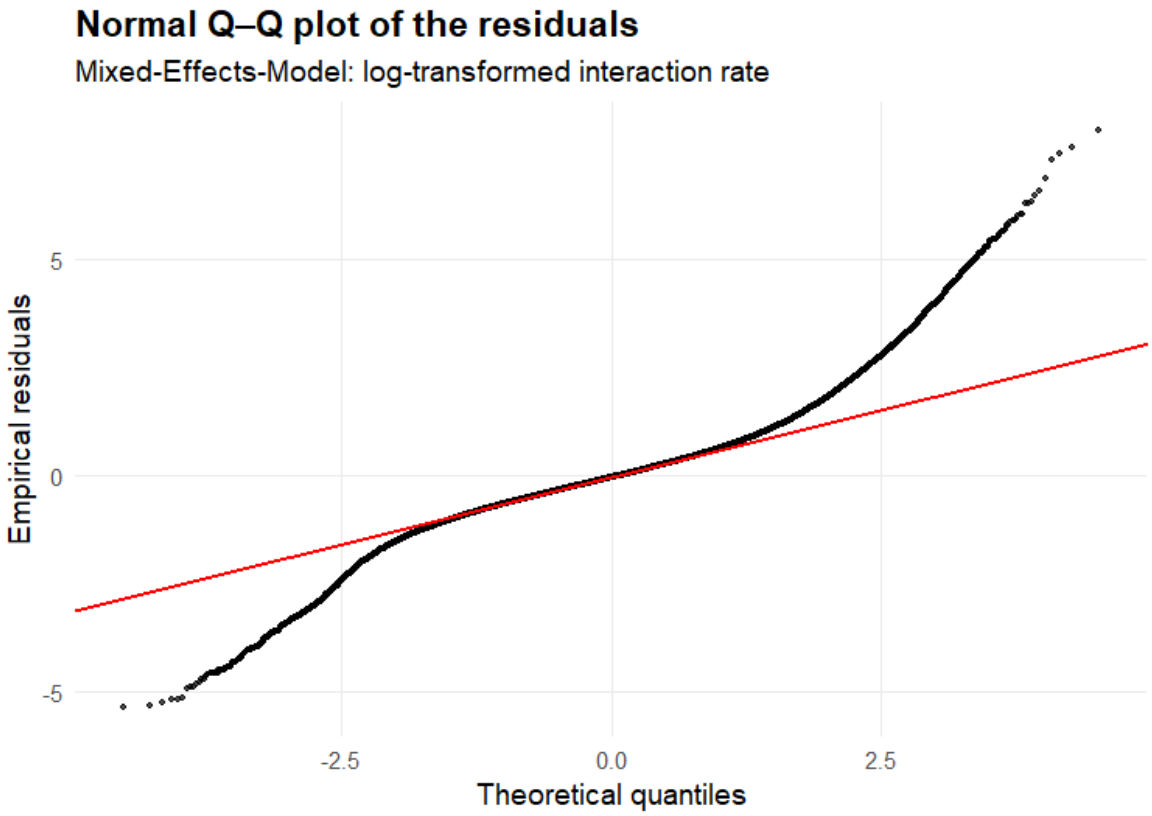



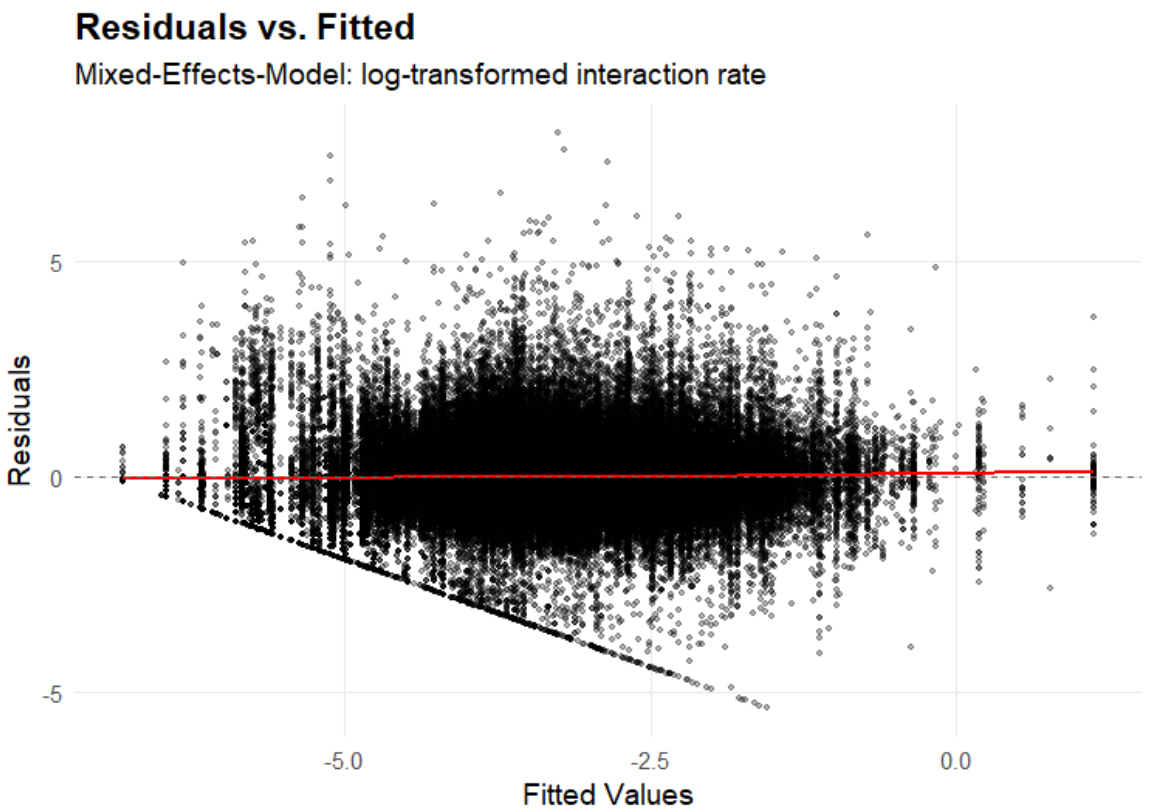
### 7.3.2 Hypothesis 3



### 7.3.3 Hypothesis 4



## 7.4 Appendix D: Code

Link to Code

## 7.5 Appendix E: Evaluation Classification US-Sample (Dataset I)

| | |
|---|---|
| Accuracy | 0,9897 |
| Precision | 0,9958 |
| Recall | 0,9897 |
| F1-Score | 0,9919 |

*Evaluation metrics of the automated classification using k-fold cross-validation (k = 5), weighted averages, authors' own calculation, n = 5 × 9,481 = 47,405.*

| **Category** | **Accuracy** | **Precision** | **Recall** | **F1-Score** |
|---|---|---|---|---|
| Person | 0,9880 | 0,2048 | 0.9670 | 0,3378 |
| Issue | 0,9817 | 0,4588 | 0,9580 | 0,6204 |
| Party | 0,9993 | 0,1233 | 0,8000 | 0,2127 |

***Figure 2:*** *Evaluation metrics of the automated classification by category using k-fold cross-validation (k = 5), authors' own calculation, n = 5 × 7,583 = 47,405.*

# 8 Data

Berg, J. (2026). *Instagram-Beiträge aus dem Bundestagswahlkampf 2025* [Datensatz]. Universität Regensburg. https://doi.org/10.5283/epub.78978

Berg, J. (2026). *Instagram-Beiträge von Prominenten aus den USA und Deutschland* [Datensatz]. Universität Regensburg. https://doi.org/10.5283/epub.78663